

LA(1)LR(0)

임기정

Abstract

This note fixes the LR(0) characteristic automaton, defines the one-symbol lookahead set attached to each completed LR(0) item, and derives the usual Read/Follow digraph computation. The proofs are written so that the essential claims appear as inclusions, fixed-point equations, and derivation chains.

Contents

1	The LR(0) Automaton	2
2	Lookahead Restriction	4
3	Read and Follow Sets	5
A	Digraph Algorithm and Implementation Specification	7
A.1	General Digraph Fixed-Point Problem	7
A.2	Instance for Read	8
A.3	Instance for Follow	9
A.4	Lookback Relation and Lookahead Construction	9
A.5	Parser-Table and Conflict Specification	9
A.6	Summary Specification	10

1 The LR(0) Automaton

Definition 1 ($\text{LRA}(G)$). Let $G \triangleq (N, T, P, S)$ be a context-free grammar. Define:

(i)

$$G' = (N', T', P', S') \triangleq (N \uplus \{S'\}, T \uplus \{\$, \}, P \cup \{[S' \rightarrow S\$]\}, S').$$

(ii)

$$V \triangleq N \uplus T, \quad V' \triangleq N' \uplus T'.$$

(iii)

$$\beta \xRightarrow{\mathbf{rm}}_{P'} \gamma \quad \xLeftrightarrow{\text{def}} \quad (\beta, \gamma) \in \{(\alpha Az, \alpha \omega z) \mid [A \rightarrow \omega] \in P', \alpha \in V'^*, z \in T'^*\}.$$

(iv)

$$I_{\text{LR}(0)} \triangleq \{[A \rightarrow \alpha \cdot \beta] \mid [A \rightarrow \alpha\beta] \in P'\}.$$

(v) For $q \subseteq I_{\text{LR}(0)}$,

$$\text{Cl}(q) =_{\mu} q \cup \{[A \rightarrow \varepsilon \cdot \omega] \mid [A \rightarrow \omega] \in P', [B \rightarrow \beta \cdot A\gamma] \in \text{Cl}(q)\}.$$

(vi)

$$q_0 \triangleq \text{Cl}(\{[S' \rightarrow \varepsilon \cdot S\$]\}).$$

(vii) For $q \subseteq I_{\text{LR}(0)}$ and $X \in V'$,

$$\text{GOTO}(q, X) \triangleq \text{Cl}(\{[A \rightarrow \alpha X \cdot \beta] \mid [A \rightarrow \alpha \cdot X\beta] \in q\}).$$

(viii)

$$Q \triangleq \text{PT} \setminus \{\emptyset\}, \quad \text{PT} =_{\mu} \{q_0\} \cup \{\text{GOTO}(q, X) \mid q \in \text{PT}, X \in V'\}.$$

(ix) The path relation is the least relation satisfying

$$\varepsilon : p \rightarrow q \iff p \in Q \wedge p = q, \quad X\alpha : p \rightarrow q \iff p \in Q \wedge \alpha : \text{GOTO}(p, X) \rightarrow q.$$

(x)

$$\mathbf{Config} \triangleq \{(\alpha : p \rightarrow q, z) \mid \alpha \in V'^*, p, q \in Q, z \in T'^*, \alpha : p \rightarrow q\}.$$

(xi)

$$\delta(q, X) \triangleq \text{GOTO}(q, X) \quad (q \in Q, X \in V').$$

(xii)

$$\mathbf{reduce}(q, t) \triangleq \{[A \rightarrow \omega] \mid [A \rightarrow \omega \cdot \varepsilon] \in q\} \quad (q \in Q, t \in T').$$

(xiii) Let $\vdash \subseteq \mathbf{Config} \times \mathbf{Config}$ be generated by:

$$\frac{q'' = \delta(q', t)}{(\alpha : q \rightarrow q', tz) \vdash (\alpha t : q \rightarrow q'', z)} \text{ Shift}$$

$$\frac{\alpha : q \rightarrow p \quad \omega : p \rightarrow q' \quad [A \rightarrow \omega] \in \mathbf{reduce}(q', t) \quad q'' = \delta(p, A)}{(\alpha \omega : q \rightarrow q', tz) \vdash (\alpha A : q \rightarrow q'', tz)} \text{ Reduce}(A \rightarrow \omega)$$

(xiv)

$$q_f \triangleq \delta(\delta(q_0, S), \$).$$

(xv) The accepted language is

$$L(\text{LRA}(G)) \triangleq \{z \in T^* \mid (\varepsilon : q_0 \rightarrow q_0, z\$) \vdash^* (S\$: q_0 \rightarrow q_f, \varepsilon)\}.$$

For $c = (\alpha : p \rightarrow q, u) \in \mathbf{Config}$, put

$$Y(c) \triangleq \alpha u \in V'^*.$$

Lemma 2 (LR(0) path invariants). *For $\eta \in V'^*$, the characteristic automaton satisfies*

$$\begin{aligned} (\exists q \in Q) \eta : q_0 \rightarrow q &\iff \eta \text{ is a viable prefix of } G', \\ S' \xRightarrow[\mathbf{rm}_{P'}]{*} \eta' Ay \xRightarrow[\mathbf{rm}_{P'}]{*} \eta' \omega y, \quad y = tz \in T'^* &\implies [A \rightarrow \omega \cdot \varepsilon] \in q \wedge [A \rightarrow \omega] \in \mathbf{reduce}(q, t). \end{aligned} \quad (1)$$

Proof. These are the standard inductive invariants of Cl, GOTO, and the path relation in Definition 1. $\square$

Lemma 3 (One-step yield invariant). *Each parser step reverses zero or one rightmost grammar step:*

$$c \vdash c' \implies \begin{cases} Y(c') = Y(c), & \text{if the step is Shift,} \\ Y(c') \xRightarrow[\mathbf{rm}_{P'}]{*} Y(c), & \text{if the step is Reduce.} \end{cases} \quad (3)$$

Proof. Indeed,

$$\begin{aligned} \text{Shift : } Y(\alpha : q \rightarrow q', tz) &= \alpha tz = Y(\alpha t : q \rightarrow q'', z), \\ \text{Reduce : } Y(\alpha A : q \rightarrow q'', tz) &= \alpha A tz \xRightarrow[\mathbf{rm}_{P'}]{*} \alpha \omega tz = Y(\alpha \omega : q \rightarrow q', tz). \end{aligned}$$

$\square$

Corollary 4 (Multi-step yield invariant). *For $c, c' \in \mathbf{Config}$,*

$$c \vdash^* c' \implies Y(c') \xRightarrow[\mathbf{rm}_{P'}]{*} Y(c). \quad (4)$$

Proof. Induct on the length of $c \vdash^* c'$ and use Lemma 3 at each parser step. $\square$

Corollary 5 (LR(0) soundness).

$$L(\mathbf{LRA}(G)) \subseteq L(G).$$

Proof.

$$\begin{aligned} z \in L(\mathbf{LRA}(G)) &\implies (\varepsilon : q_0 \rightarrow q_0, z\$) \vdash^* (S\$: q_0 \rightarrow q_f, \varepsilon) \\ &\xRightarrow[\mathbf{rm}_{P'}]{*} S\$ \xRightarrow[\mathbf{rm}_{P'}]{*} z\$ \\ &\implies S \Rightarrow_P^* z \\ &\implies z \in L(G). \end{aligned}$$

The third implication deletes the endmarker: S' never occurs on a right-hand side, so $[S' \rightarrow S\$]$ is unused in a derivation from $S\$$, and $\$$ is never rewritten. $\square$

Lemma 6 (LR(0) completeness).

$$L(G) \subseteq L(\mathbf{LRA}(G)).$$

Proof. Let $z \in L(G)$, so $S \Rightarrow_P^* z$. Choose a rightmost derivation in G' :

$$S\$ = \gamma_0 \xRightarrow[\mathbf{rm}_{P'}]{*} \gamma_1 \xRightarrow[\mathbf{rm}_{P'}]{*} \cdots \xRightarrow[\mathbf{rm}_{P'}]{*} \gamma_N = z\$. \quad (5)$$

Write its k -th step as

$$\gamma_k = \alpha_k A_k y_k \xRightarrow[\mathbf{rm}_{P'}]{*} \alpha_k \omega_k y_k = \gamma_{k+1}, \quad [A_k \rightarrow \omega_k] \in P', \quad y_k = t_k z_k \in T'^*. \quad (6)$$

By (2), for each k there exist $p_k, q_k \in Q$ such that

$$\alpha_k : q_0 \rightarrow p_k, \quad \omega_k : p_k \rightarrow q_k, \quad [A_k \rightarrow \omega_k \cdot \varepsilon] \in q_k, \quad [A_k \rightarrow \omega_k] \in \mathbf{reduce}(q_k, t_k). \quad (7)$$

Reading (5) from right to left gives, for $k = N - 1, \dots, 0$,

$$(\alpha_k \omega_k : q_0 \rightarrow q_k, y_k) \vdash (\alpha_k A_k : q_0 \rightarrow \delta(p_k, A_k), y_k), \quad (8)$$

with any intervening terminal symbols consumed by Shift steps along the viable prefix path. Iterating (8) and then shifting the endmarker yields

$$(\varepsilon : q_0 \rightarrow q_0, z\$) \vdash^* (S\$: q_0 \rightarrow q_f, \varepsilon), \quad q_f = \delta(\delta(q_0, S), \$).$$

Hence $z \in L(\mathbf{LRA}(G))$. □

Corollary 7 (Correctness of $\mathbf{LRA}(G)$).

$$L(G) = L(\mathbf{LRA}(G)), \quad L(G) \triangleq \{z \in T^* \mid S \Rightarrow_P^* z\}.$$

Proof. Corollary 5 and Lemma 6 give

$$L(\mathbf{LRA}(G)) \subseteq L(G), \quad L(G) \subseteq L(\mathbf{LRA}(G)).$$

□

2 Lookahead Restriction

Define

$$\mathbf{LA} : \{(q, [A \rightarrow \omega]) \mid q \in Q, [A \rightarrow \omega \cdot \varepsilon] \in q\} \longrightarrow \mathcal{P}(T')$$

by

$$\mathbf{LA}(q, [A \rightarrow \omega]) \triangleq \left\{ t \in T' \mid S' \xRightarrow[\mathbf{rm}_P^*]{\alpha A t z} \alpha \omega : q_0 \rightarrow q, \alpha \in V'^*, z \in T'^* \right\}. \quad (9)$$

Replace the $\mathbf{LR}(0)$ reduction function by

$$\mathbf{reduce}_{\mathbf{LA}}(q, t) \triangleq \{[A \rightarrow \omega] \mid [A \rightarrow \omega \cdot \varepsilon] \in q, t \in \mathbf{LA}(q, [A \rightarrow \omega])\}. \quad (10)$$

Let $\vdash_{\mathbf{LA}}$ be the transition relation obtained from $\vdash$ by replacing $\mathbf{reduce}$ with $\mathbf{reduce}_{\mathbf{LA}}$, and let $L(\vdash_{\mathbf{LA}})$ be its accepted language.

Lemma 8 (Guarded soundness).

$$L(\vdash_{\mathbf{LA}}) \subseteq L(G).$$

Proof. From (10),

$$\mathbf{reduce}_{\mathbf{LA}}(q, t) \subseteq \mathbf{reduce}(q, t) \implies \vdash_{\mathbf{LA}} \subseteq \vdash \implies \vdash_{\mathbf{LA}}^* \subseteq \vdash^*.$$

Therefore

$$L(\vdash_{\mathbf{LA}}) \subseteq L(\mathbf{LRA}(G)) = L(G) \quad \text{by coro 7.} \quad (11)$$

□

Lemma 9 (Guarded completeness).

$$L(G) \subseteq L(\vdash_{\mathbf{LA}}).$$

Proof. Let $z \in L(G)$. In the completeness construction of corollary 7, every Reduce step has the form

$$(\alpha\omega : q_0 \rightarrow q, tz') \vdash (\alpha A : q_0 \rightarrow \delta(p, A), tz'),$$

where

$$S' \xrightarrow{\text{rm}}_{P'} S \xrightarrow{\text{rm}}_{P'}^* \alpha Atz', \quad \alpha\omega : q_0 \rightarrow q, \quad [A \rightarrow \omega \cdot \varepsilon] \in q.$$

By (9),

$$t \in \mathbf{LA}(q, [A \rightarrow \omega]) \iff [A \rightarrow \omega] \in \mathbf{reduce}_{\mathbf{LA}}(q, t).$$

Thus every Reduce step used in that accepting LR(0) computation is also legal for $\vdash_{\mathbf{LA}}$; Shift steps are unchanged. Hence

$$L(G) \subseteq L(\vdash_{\mathbf{LA}}).$$

□

Corollary 10 (Language preservation).

$$L(\vdash_{\mathbf{LA}}) = L(G).$$

Proof. Combine Lemmas 8 and 9. □

Lemma 11 (Conflict-free table condition). *If the resulting table has no shift/reduce or reduce/reduce conflict, its action is single-valued under one-symbol lookahead, hence it is an LALR(1) table.*

Proof. The action at (q, t) consists of the shift $\delta(q, t)$ when it is nonempty and the reductions in $\mathbf{reduce}_{\mathbf{LA}}(q, t)$. Absence of shift/reduce and reduce/reduce conflicts means that this action is single-valued under one-symbol lookahead, which is precisely the LALR(1) table condition. □

Theorem 12 (Lookahead-guarded reductions).

If the resulting table has no shift/reduce or reduce/reduce conflict, it is an LALR(1) parser and accepts $L(G)$.

Proof. Corollary 10 gives acceptance of $L(G)$, and Lemma 11 gives the LALR(1) table condition under the stated absence of conflicts. □

3 Read and Follow Sets

Let

$$D \triangleq \{(p, A) \mid p \in Q, A \in N', \delta(p, A) \neq \emptyset\}.$$

For a relation R , write

$$R!x \triangleq \{y \mid (x, y) \in R\}.$$

Define **Read**, **Follow** $\subseteq D \times T'$ as the least relations generated by the rules

$$\frac{\delta(\delta(p, A), t) \neq \emptyset}{t \in \mathbf{Read}!(p, A)} \text{DR} \quad \frac{\delta(p, A) = r \quad C \Rightarrow_{P'}^* \varepsilon \quad (r, C) \in D}{\mathbf{Read}!(r, C) \subseteq \mathbf{Read}!(p, A)} \text{reads}$$

$$\frac{t \in \mathbf{Read}!(p, A)}{t \in \mathbf{Follow}!(p, A)} \text{Read} \quad \frac{[B \rightarrow \beta \cdot A\gamma] \in p \quad \beta : p' \rightarrow p \quad \gamma \Rightarrow_{P'}^* \varepsilon \quad (p', B) \in D}{\mathbf{Follow}!(p', B) \subseteq \mathbf{Follow}!(p, A)} \text{includes}$$

Define the semantic read and follow sets on D :

$$\mathbf{Read}(p, A) \triangleq \{t \in T' \mid \exists r, s, \gamma. r = \delta(p, A) \wedge \gamma \Rightarrow_{P'}^* \varepsilon \wedge \gamma t : r \rightarrow s\}, \quad (12)$$

$$\mathbf{Follow}(p, A) \triangleq \left\{ t \in T' \mid \exists \alpha, z. S' \xrightarrow{\text{rm}}_{P'}^* \alpha Atz \wedge \alpha : q_0 \rightarrow p \wedge z \in T'^* \right\}. \quad (13)$$

For $(p, A) \in D$, set

$$\text{DR}(p, A) \triangleq \{t \in T' \mid \delta(\delta(p, A), t) \neq \emptyset\}.$$

Let

$$\Phi_R(F)(p, A) = \text{DR}(p, A) \cup \bigcup_{\substack{\delta(p, A)=r \\ C \Rightarrow_{P'}^* \varepsilon \\ (r, C) \in D}} F(r, C).$$

Lemma 13 (Semantic Read fixed point). *For $(p, A) \in D$,*

$$\mathbf{Read}!(p, A) = \mu \Phi_R(p, A) = \text{Read}(p, A). \quad (14)$$

Proof. The two **Read**-rules define the least fixed point of the monotone operator Φ_R . By (12), $t \in \text{Read}(p, A)$ iff for some nullable $\gamma = C_1 \cdots C_n \in N'^*$,

$$\delta(p, A) = r_0, \quad \delta(r_{i-1}, C_i) = r_i \ (1 \leq i \leq n), \quad \delta(r_n, t) \neq \emptyset.$$

Equivalently,

$$t \in \underbrace{\Phi_R^n(\emptyset)(p, A)}_{n \text{ reads edges, then DR}},$$

and hence (14). □

Let

$$\Phi_F(F)(p, A) = \text{Read}(p, A) \cup \bigcup_{\substack{[B \rightarrow \beta \cdot A \gamma] \in p \\ \beta : p' \rightarrow p \\ \gamma \Rightarrow_{P'}^* \varepsilon \\ (p', B) \in D}} F(p', B).$$

Lemma 14 (Semantic Follow fixed point). *For $(p, A) \in D$,*

$$\mathbf{Follow}!(p, A) = \mu \Phi_F(p, A) = \text{Follow}(p, A). \quad (15)$$

Proof. By Lemma 13, the two **Follow**-rules define the least fixed point of Φ_F . First Follow is closed under Φ_F . For the Read-seed:

$$\begin{aligned} t \in \text{Read}(p, A) &\implies \exists \alpha, \gamma, z. \alpha : q_0 \rightarrow p \wedge \gamma \Rightarrow_{P'}^* \varepsilon \wedge S' \xRightarrow{\mathbf{rm}_{P'}}^* \alpha A \gamma t z \\ &\implies S' \xRightarrow{\mathbf{rm}_{P'}}^* \alpha A \gamma t z \xRightarrow{\mathbf{rm}_{P'}}^* \alpha A t z \\ &\implies t \in \text{Follow}(p, A). \end{aligned}$$

For an includes edge,

$$\begin{aligned} t \in \text{Follow}(p', B), \quad [B \rightarrow \beta \cdot A \gamma] \in p, \quad \beta : p' \rightarrow p, \quad \gamma \Rightarrow_{P'}^* \varepsilon \Big\} &\implies S' \xRightarrow{\mathbf{rm}_{P'}}^* \alpha' B t z \\ &\xRightarrow{\mathbf{rm}_{P'}}^* \alpha' \beta A \gamma t z \\ &\xRightarrow{\mathbf{rm}_{P'}}^* \alpha' \beta A t z, \end{aligned}$$

with $\alpha' \beta : q_0 \rightarrow p$; hence

$$\text{Follow}(p', B) \subseteq \text{Follow}(p, A).$$

Conversely, let $t \in \text{Follow}(p, A)$, witnessed by

$$S' \xRightarrow{\mathbf{rm}_{P'}}^* \alpha A t z, \quad \alpha : q_0 \rightarrow p.$$

Looking at the production that created this occurrence of A gives a coroor

$$S' \xRightarrow{\mathbf{rm}_{P'}}^* \alpha' B t' z' \xRightarrow{\mathbf{rm}_{P'}}^* \alpha' \beta A \gamma t' z' \xRightarrow{\mathbf{rm}_{P'}}^* \alpha A t z, \quad \alpha = \alpha' \beta.$$

Thus exactly one of the following alternatives holds:

$$\begin{cases} \gamma = \gamma_0 t \gamma_1, & \gamma_0 \Rightarrow_{P'}^* \varepsilon, \quad t \in \text{Read}(p, A), \\ \gamma \Rightarrow_{P'}^* \varepsilon, & t \in \text{Follow}(p', B) \text{ and } (p, A) \rightsquigarrow_{\text{includes}} (p', B). \end{cases}$$

In the second case the witnessing rightmost derivation is strictly shorter. Induction on that length gives

$$\text{Follow}(p, A) \subseteq \mu\Phi_F(p, A).$$

Together with closure, this proves (15). $\square$

Corollary 15 (Lookahead by Follow). *Whenever $q \in Q$ and $[A \rightarrow \omega \cdot \varepsilon] \in q$,*

$$\mathbf{LA}(q, [A \rightarrow \omega]) = \{t \in T' \mid \exists p \in Q. \omega : p \rightarrow q \wedge \delta(p, A) \neq \emptyset \wedge t \in \mathbf{Follow}!(p, A)\}. \quad (16)$$

Proof. For $q \in Q$ and $[A \rightarrow \omega \cdot \varepsilon] \in q$,

$$\begin{aligned} t \in \mathbf{LA}(q, [A \rightarrow \omega]) &\iff \exists \alpha, z. S' \xrightarrow[\text{rm}_{P'}^*]{\alpha} \alpha A t z \wedge \alpha \omega : q_0 \rightarrow q \\ &\iff \exists p, \alpha, z. S' \xrightarrow[\text{rm}_{P'}^*]{\alpha} \alpha A t z \wedge \alpha : q_0 \rightarrow p \wedge \omega : p \rightarrow q \\ &\iff \exists p. \omega : p \rightarrow q \wedge \delta(p, A) \neq \emptyset \wedge t \in \mathbf{Follow}(p, A) \\ &\stackrel{(15)}{\iff} \exists p. \omega : p \rightarrow q \wedge \delta(p, A) \neq \emptyset \wedge t \in \mathbf{Follow}!(p, A). \end{aligned}$$

The side condition $\delta(p, A) \neq \emptyset$ is automatic for $A \neq S'$: tracing $[A \rightarrow \omega \cdot \varepsilon] \in q$ backward along $\omega : p \rightarrow q$ gives $[A \rightarrow \varepsilon \cdot \omega] \in p$, which is introduced by an item with the dot before A . For $A = S'$, both sides are empty because S' occurs on no right-hand side and the parser does not reduce $[S' \rightarrow S\$]$. This is (16). $\square$

Theorem 16 (Digraph computation of lookahead). *The least relations **Read**, **Follow** generated above compute every lookahead set by the formula (16).*

Proof. The Read and Follow fixed-point claims are Lemmas 13 and 14; applying them to the semantic definition of **LA** gives Corollary 15, which is the displayed formula of the theorem. $\square$

A Digraph Algorithm and Implementation Specification

This appendix gives an executable specification for computing the sets used in Theorem 16. The construction is the usual digraph view of LALR(1) lookahead computation: direct seed sets are placed on nodes, dependency edges describe set inclusions, and the least fixed point is obtained by collapsing strongly connected components and propagating set unions along the resulting DAG [1, 2, 3].

A.1 General Digraph Fixed-Point Problem

Let X be a finite set of nodes, let $E \subseteq X \times X$ be a directed edge relation, and let $\text{Seed} : X \rightarrow \mathcal{P}(T')$. We use the edge orientation

$$x \rightsquigarrow y \quad \text{to mean} \quad F(y) \subseteq F(x).$$

Thus the desired solution is the least function $F : X \rightarrow \mathcal{P}(T')$ satisfying

$$F(x) = \text{Seed}(x) \cup \bigcup_{x \rightsquigarrow y} F(y). \quad (17)$$

Algorithm DIGRAPH(X, E, Seed).

1. Compute the strongly connected components of (X, E) , for example by Tarjan's depth-first-search algorithm.
2. Build the condensation DAG $\mathcal{C}$. Its nodes are SCCs, and $C \rightsquigarrow D$ is an edge when some $x \in C$ and $y \in D$ satisfy $x \rightsquigarrow y$, with $C \neq D$.
3. Initialize, for every SCC C ,

$$\text{Val}(C) \leftarrow \bigcup_{x \in C} \text{Seed}(x).$$

4. Process the SCCs in reverse topological order with respect to $\rightsquigarrow$, so every successor D of C has already been processed, and put

$$\text{Val}(C) \leftarrow \text{Val}(C) \cup \bigcup_{C \rightsquigarrow D} \text{Val}(D).$$

5. Return $F(x) \triangleq \text{Val}(C_x)$, where C_x is the SCC containing x .

Correctness condition. For every SCC C , after step 4,

$$\text{Val}(C) = \bigcup \{ \text{Seed}(y) \mid \text{some } x \in C \text{ reaches } y \text{ in } (X, E) \}.$$

Therefore the returned F is exactly the least solution of (17). This is the only property of the graph algorithm needed by Theorem 16.

Complexity. Let $b = \lceil |T'|/w \rceil$, where w is the machine word size used for bitsets. SCC construction is $O(|X| + |E|)$, and set propagation is

$$O((|X| + |E|)b).$$

A simple work-list implementation that repeatedly applies

$$F(x) := F(x) \cup F(y) \quad (x \rightsquigarrow y)$$

is also correct, but the SCC version avoids repeated propagation inside cycles.

A.2 Instance for Read

First compute nullable nonterminals by the least fixed point

$$\text{Null}(A) \iff \exists [A \rightarrow X_1 \cdots X_k] \in P' \text{ such that every } X_i \text{ is nullable,}$$

where terminals are never nullable and $k = 0$ is allowed. Extend this to strings by

$$\text{Null}(X_1 \cdots X_k) \iff \bigwedge_{i=1}^k \text{Null}(X_i).$$

For $x = (p, A) \in D$, define the direct-read seed

$$\text{DR}(p, A) \triangleq \{t \in T' \mid \delta(\delta(p, A), t) \neq \emptyset\}.$$

Define $E_R \subseteq D \times D$ by

$$(p, A) \rightsquigarrow_R (r, C) \iff \delta(p, A) = r \wedge C \in N' \wedge \text{Null}(C) \wedge (r, C) \in D.$$

If $(r, C) \notin D$, then **Read**! $(r, C) = \emptyset$, so no edge is needed. Compute

$$\mathbf{Read}! (p, A) \triangleq \text{DIGRAPH}(D, E_R, \text{DR})(p, A).$$

A.3 Instance for Follow

The seed for follow propagation is the already computed read set:

$$\text{Seed}_F(p, A) \triangleq \mathbf{Read}!(p, A).$$

Define $E_I \subseteq D \times D$ by

$$(p, A) \rightsquigarrow_I (p', B) \iff [B \rightarrow \beta \cdot A\gamma] \in p \wedge \beta : p' \rightarrow p \wedge \text{Null}(\gamma) \wedge (p', B) \in D.$$

The orientation means

$$\mathbf{Follow}!(p', B) \subseteq \mathbf{Follow}!(p, A).$$

Compute

$$\mathbf{Follow}!(p, A) \triangleq \text{DIGRAPH}(D, E_I, \text{Seed}_F)(p, A).$$

A.4 Lookback Relation and Lookahead Construction

For every completed item $[A \rightarrow \omega \cdot \varepsilon] \in q$, define

$$\text{LB}(q, A \rightarrow \omega) \triangleq \{(p, A) \in D \mid \omega : p \rightarrow q\}.$$

Then

$$\mathbf{LA}(q, [A \rightarrow \omega]) \triangleq \bigcup_{(p, A) \in \text{LB}(q, A \rightarrow \omega)} \mathbf{Follow}!(p, A),$$

which is (16) written as an implementation step.

Implementation note. The query $\omega : p \rightarrow q$ can be implemented by following δ -edges from p along ω . The query $\beta : p' \rightarrow p$ in E_I can be implemented by caching reverse-path queries keyed by (β, p) , or by storing predecessor edges during construction of the $\text{LR}(0)$ automaton.

A.5 Parser-Table and Conflict Specification

After $\mathbf{LA}$ has been computed, construct actions as follows.

1. If $\delta(q, t) \neq \emptyset$ for $t \in T'$, add the shift action

$$q \xrightarrow{t} \delta(q, t).$$

2. For each $[A \rightarrow \omega \cdot \varepsilon] \in q$ and each $t \in \mathbf{LA}(q, [A \rightarrow \omega])$, add the reduce action $A \rightarrow \omega$ under lookahead t .
3. Report a shift/reduce conflict when both a shift action and a reduce action are present for the same pair (q, t) .
4. Report a reduce/reduce conflict when two distinct reduce productions are present for the same pair (q, t) .

The accepting condition remains the one in Definition 1: the parser accepts when it reaches

$$(S\$: q_0 \rightarrow q_f, \varepsilon).$$

Equivalently, in a conventional ACTION/GOTO table, q_f may be treated as the accepting state after the input marker has been consumed.

A.6 Summary Specification

The complete computation required by Theorem 16 is:

1. Build G' , Q , and δ as in Definition 1.
2. Compute nullable nonterminals and nullable strings.
3. Build D , DR , and E_R ; compute **Read** by $\text{DIGRAPH}(D, E_R, DR)$.
4. Build E_I ; compute **Follow** by $\text{DIGRAPH}(D, E_I, \text{Read})$.
5. Build LB and compute each $\mathbf{LA}(q, [A \rightarrow \omega])$ by unioning the appropriate **Follow**-sets.
6. Construct the guarded reduce actions and check conflicts.

The postcondition is that the computed **Read**, **Follow**, and **LA** are the least sets satisfying Theorem 16, and hence the lookahead-restricted parser of Theorem 12 uses exactly the LALR(1) lookahead sets determined by the LR(0) automaton.

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